Adiabatic flame temperature

CLASS 4

Learning Outcomes

Definition of the Adiabatic flame temperature concept

Understanding of the important variables affecting the process

Illustration of the methods for adiabatic flame temperature calculation

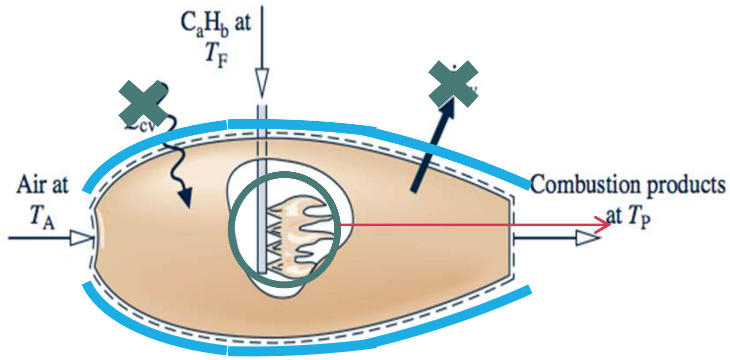
Adiabatic flame temperature

In a open system energy is transferred through:

- Mass, Work and Heat.

If W_{syst}=0, only through products and heat trans

- Therefore:
  - Smaller Q, higher T_{prod}



Adiabatic flame temperature:

- In an adiabatic system:
  - Product temperature reaches maximum
  - Energy is only used to heat the products.
  - Also known as Adiabatic combustion

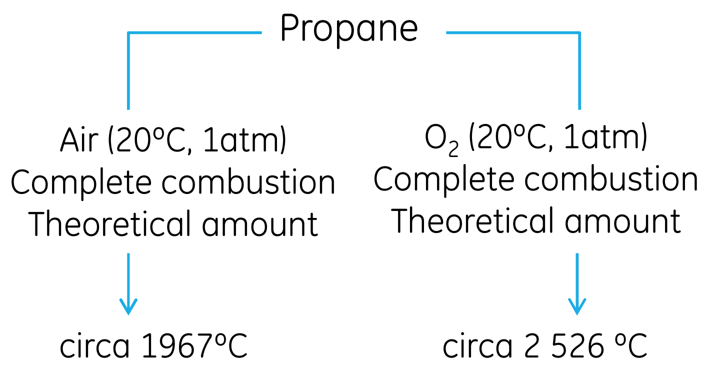
Adiabatic flame temperature

The T_{ad} is specific for a process, it depends of:

- The type of fuel
- Temperature and pressure of the reactants
- Rate of combustion – complete or incomplete
- Amount of air/oxidant used.

Maximum Tad ($T_{ad,max}$) value <u>for a specific process</u>:

- Complete combustion with theoretical amount of air (ideal)
  - Specific process: certain fuel, with reactants at certain state.
- The real measured Tad value is several degrees below $T_{ad,max}$
  - Why do you think this happens?



What affects the max t_{ad} value?

Heat losses occur – no system is 100% adiabatic:

- Conduction, radiation and convection

Excess of air– there is an adequate amount

- Functions as a diluent.
  - Also, ideally, nitrogen doesn't react but its heated
- High excess aids dissociation processes

Dissociation:

- At high temperatures molecules break down
- At low T – CO_2, H_2O, N_2 At high T – CO, H_2, ^-O, H^-, ^-OH

State of reactants

- Heating and phase changes implies energy consumption

Important variables

AF$_{ratio}$/ FA$_{ratio}$:

- The combustion is complete or incomplete

Temperature air

- Increase in Tair, increases Tad
  - No energy losses from heating.

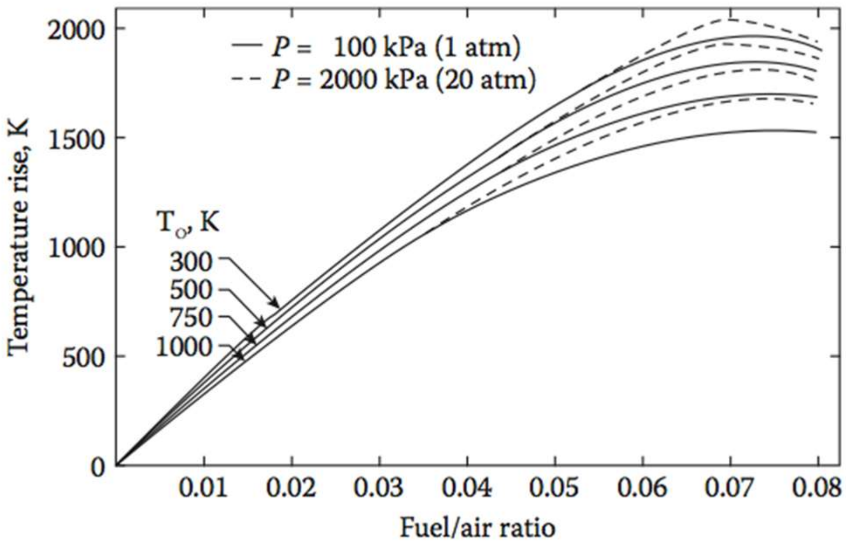
Pressure

- For a constant Tair, increase pressure increase Tad
  - High T, increases P and degree of dissociation of products
  - This has an effect over system volume
  - Constant volume, pressure increases, dissociation reduces

Air vitiation

- Precombustion: CO_2 and H_2O

Adiabatic temperature rise curve for JP5 fuel.



Importance of Tad

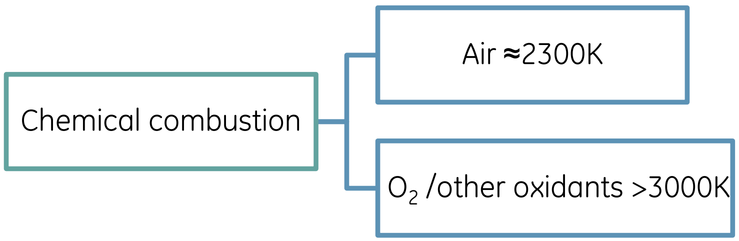
Maximum process temperature:

- Limits the process
  - Rarely a real process reaches this temperature
- Calculates pollutants amount

Estimates thermal effects:

- Considerations for materials
- Considerations in design, handle and security of process

Important for combustion efficiency.



Determining Tad

Calculate $H_{reactants}$

- Quick and direct
- Their state is known
- Data can be found in tables

Determining $H_{products}$ is different

- Tprod = Tad, is unknown

Can be calculated by:

- Enthalpy balance iteration
- Average specific heat (Cp)

$$\mathrm{Q} - \mathrm{W} = \mathrm{H}$$

If, there is no work done
And Ek, Ep negligible

$$\mathrm{Q} = \sum (N_i \bar{h}_i)_R - \sum (N_i \bar{h}_i)_P$$

$$\sum (N_i \bar{h}_i)_P = \sum (N_i \bar{h}_i)_R$$

$$\sum N_P (\bar{h}_f^\circ + \bar{h}(T) - \bar{h}^\circ)_P = \sum N_R (\bar{h}_f^\circ + \bar{h}(T) - \bar{h}^\circ)_R$$

Method 1: Enthalpy balance

General equation for open system:

$$\boxed{\sum N_P(\bar{h}_f^\circ + \bar{h}(T_{ad}) - \bar{h}^\circ)_P = \sum N_R(\bar{h}_f^\circ + \bar{h}(T_F) - \bar{h}^\circ)_R}$$

T_F of the products $\rightarrow T_{ad}$

Re-arranging the terms:

$$\sum N_P\left(\bar{h}(T_{ad}) - \bar{h}^\circ\right)_P = \sum N_R \bar{h}_{f_R}^\circ - \sum N_P \bar{h}_{f_P}^\circ + \sum N_R\left(\bar{h}(T_F) - \bar{h}^\circ\right)_R$$

$$\boxed{\sum N_P\left(\bar{h}(T_{ad}) - \bar{h}^\circ\right)_P = -\mathrm{q}_{rsn} + \sum N_R\left(\bar{h}(T_F) - \bar{h}^\circ\right)_R}$$

Depending of the scenario, we have to make supposition on the temperature we are calculating

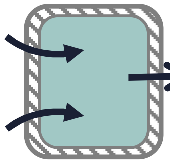
$$\sum N_P(\bar{h}_f^\circ + \bar{h}(T_{guess}) - \bar{h}^\circ)_P = \sum N_R(\bar{h}_f^\circ + \bar{h}(T_{guess}) - \bar{h}^\circ)_R$$

We will iterate with the unknown temperature until the energy balance is satisfied:

$$\sum N_P H_P = \sum N_R H_R$$

Estimate the adiabatic flame temperature at constant pressure for a stoichiometric combustion of octane with air, if the reactants are at 298K, 1 atm.

Suppositions: 1) open system, steady-state; 2)No works is done; 3) All gases are ideal gases



Chemical reaction balanced:

$$C_8H_{18} + 12.5(O_2 + 3.76N_2) \rightarrow 8CO_2 + 9H_2O + 47N_2$$

Energy balance:

$$\sum N_P \bar{h}_p(T_{ad}) = \sum N_R(\bar{h}_f^{\circ})_R - \sum N_P(\bar{h}_f^{\circ})_P + \sum N_P(\bar{h}^{\circ})_P$$

$$\sum \mathrm{N_R}(\bar{\mathrm{h}}_\mathrm{f}^{\circ})_\mathrm{R} = -249950 \text{ kJ/kmol}$$

$$\sum \mathrm{N_P}(\bar{\mathrm{h}}_\mathrm{f}^{\circ})_\mathrm{P} = -5\,324\,540 \text{ kJ/kmol}$$

$$\sum \mathrm{N_P}(\bar{\mathrm{h}}^{\circ})_\mathrm{P} = 571\,491 \text{ kJ/kmol}$$

Enthalpy of formation

	h_f°(kJ/kmol)
C_8H_{18}	-249 950
O_2	0
N_2	0
$H_2O_{(g)}$	-241 820
CO_2	-393 529

In order to calculate the sensible enthalpy of the products, a final temperature of the outlet must be known – or supposed.

Supposed T = 2500 ºK

$$\sum N_P(\bar{h}(T_{2500K}))_P = 5\ 930\ 239\ \text{kJ/kmol}$$

$$5\ 930\ 239\ \text{kJ/kmol} = 5\ 646\ 081\ \text{kJ/kmol}$$

Our guess give us a higher value

Supposed T = 2000 ºK

$$\sum N_P(\bar{h}(T_{2000K}))_P = 4\ 595\ 839\ \text{kJ/kmol}$$

$$4\ 595\ 839\ \text{kJ/kmol} = 5\ 646\ 081\ \text{kJ/kmol}$$

Our guess give us a lower value

Extrapolate or interpolate:

Give us a Tad = 2 392.87ºK

H at different temperatures (kJ/kmol)

T (ºK)	CO_2	H_2O	N_2
298	9364	9904	8669
2000	100804	82593	64810
2200	112939	92940	72040
2500	131290	108868	82981
2700	143620	119717	90328
3000	162226	136264	101407

Method 2: $Cp_{i,avg}$

It's also based in a energy and mass balance

$$\sum N_P\left(\bar{h}(T_{ad}) - \bar{h}^{\circ}\right)_P = \sum N_R \bar{h}^{\circ}_{f\,R} - \sum N_P \bar{h}^{\circ}_{f\,P} + \sum N_R\left(\bar{h}(T_F) - \bar{h}^{\circ}\right)_R$$

Based on the supposition that we can approximate the term of sensible enthalpy with an average specific heat function:

$$\sum N_P\left(\bar{h}(T_{ad}) - \bar{h}^{\circ}\right)_P \cong (T_{ad} - T_0)\sum N_p \overline{Cp}_{avg,p}$$

$$(T_{ad} - T_0)\sum N_p \overline{Cp}_{avg,p} = \sum N_R \bar{h}^{\circ}_{f\,R} - \sum N_P \bar{h}^{\circ}_{f\,P} + \sum N_R\left(\bar{h}(T_F) - \bar{h}^{\circ}\right)_R$$

We can solve for our unknown variable:

$$\boxed{T_{ad} = T_0 + \frac{\sum N_R \bar{h}^{\circ}_{f\,R} - \sum N_P \bar{h}^{\circ}_{f\,P} + \sum N_R\left(\bar{h}(T_F) - \bar{h}^{\circ}\right)_R}{\sum N_p \overline{Cp}_{avg,p}}}$$

- We have the variable average specific heat

$$\overline{Cp}_{avg,P} \text{ is } \overline{Cp}_P \text{ @ } \bar{T} \qquad \bar{T} = AVG(T_{prod}, T_{react})$$ Why?

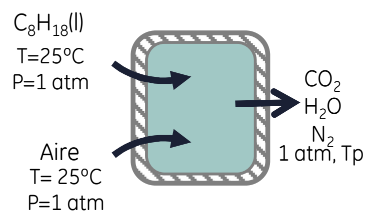
  - There are tabulated data for specific heat at different temperatures and for a variety of compounds.

Method 2 Data for $Cp_{i,avg}$

T	CO2cp	H2Ocp	O2cp	N2cp	COcp
298	37.2	33.45	29.31	29.07	29.07
300	37.28	33.47	29.33	29.07	29.08
400	41.27	34.44	30.21	29.32	29.43
500	44.57	35.34	31.11	29.63	29.86
600	47.31	36.29	32.03	30.08	30.41
700	49.61	37.36	32.93	30.68	31.09
800	51.55	38.59	33.76	31.39	31.86
900	53.13	39.93	34.45	32.13	32.63
1000	54.36	41.31	34.93	32.76	33.25
1200	56.2	43.87	35.59	33.71	34.15
1400	57.67	46.1	36.2	34.48	34.87
1600	58.83	48.03	36.77	35.1	35.45
1800	59.73	49.7	37.29	35.59	35.91
2000	60.43	51.14	37.79	35.99	36.27
2200	60.96	52.38	38.25	36.3	36.55
2400	61.37	53.43	38.68	36.54	36.77
2600	61.7	54.34	39.09	36.74	36.95
2800	61.96	55.11	39.48	36.89	37.09
3000	62.19	55.78	39.84	37.03	37.21
3200	62.4	56.35	40.19	37.14	37.32
3400	62.61	56.85	40.52	37.25	37.42
3500	62.72	57.07	40.68	37.3	37.47

Estimate the adiabatic flame temperature at constant pressure for a stoichiometric combustion of octane with air, if the reactants are at 298K, 1 atm

Suppositions: 1) open system, steady-state; 2)No works is done; 3) All gases are ideal gases



Chemical reaction balanced:

$$C_8H_{18} + 12.5(O_2 + 3.76N_2) \rightarrow 8CO_2 + 9H_2O + 47N_2$$

Based on the energy balance:

$$T_{ad} = T_0 + \frac{\sum N_R \bar{h}^{\circ}_{f\,R} - \sum N_P \bar{h}^{\circ}_{f\,P} + \sum N_R\left(\bar{h}(T_F) - \bar{h}^{\circ}\right)_R}{\sum N_p \overline{Cp}_{avg,p}}$$

$$\sum N_R(\bar{h}^{\circ}_f)_R = -249950 \text{ kJ/kmol}$$

$$\sum N_P(\bar{h}^{\circ}_f)_P = -5\,324\,540 \text{ kJ/kmol}$$

$$\sum N_R\left(\bar{h}(T_F) - \bar{h}^{\circ}\right)_R = 0$$

Calculating Cp_{avg}, @ T_{avg}:

1. Suppose a T_{ad} value $\quad T_{ad,guess} = 3\,000°K$
2. Obtain T_{avg} $\quad \bar{T} = 0.5(3000 + 298) = 1649 \text{ K} \sim 1650 \text{ K}$
3. Obtain $Cp_{avg,p}$ @ T_{avg} from table of Cp

$$\overline{Cp}_{CO_2,1650K} = 59.08 \text{ kJ/kmol}\cdot\text{K}$$

$$\overline{Cp}_{H_2O,1650K} = 48.47 \text{ kJ/kmol}\cdot\text{K}$$

$$\overline{Cp}_{N_2,1650K} = 35.23 \text{ kJ/kmol}\cdot\text{K}$$

T	CO2cp	H2Ocp	O2cp	N2cp
298	37.2	33.45	29.31	29.07
300	37.28	33.47	29.33	29.07
400	41.27	34.44	30.21	29.32
500	44.57	35.34	31.11	29.63
600	47.31	36.29	32.03	30.08
700	49.61	37.36	32.93	30.68
800	51.55	38.59	33.76	31.39
900	53.13	39.93	34.45	32.13
1000	54.36	41.31	34.93	32.76
1200	56.2	43.87	35.59	33.71
1400	57.67	46.1	36.2	34.48
1600	58.83	48.03	36.77	35.1
1800	59.73	49.7	37.29	35.59
2000	60.43	51.14	37.79	35.99

1

$T_{ad,guess} = 3000\ K$

$T_{avg} = 1649\ K$

$T_{ad,calc} = 2276.74\ K$

$C_{P_{CO_2}} = 59.072\ kJ/kmol \cdot K$

$C_{P_{H_2o}} = 48.462\ kJ/kmol \cdot K$

$C_{P_{N_2}} = 35.230\ kJ/kmol \cdot K$

2

$T_{ad,guess} = 2276.74\ K$

$T_{avg} = 1287.37\ K$

$T_{ad,calc} = 2360.71\ K$

$C_{P_{CO_2}} = 56.884\ kJ/kmol \cdot K$

$C_{P_{H_2o}} = 44.882\ kJ/kmol \cdot K$

$C_{P_{N_2}} = 34.066\ kJ/kmol \cdot K$

3

$T_{ad,guess} = 2360.71\ K$

$T_{avg} = 1329.35\ K$

$T_{ad,calc} = 2348.90\ K$

$C_{P_{CO_2}} = 57.189\ kJ/kmol \cdot K$

$C_{P_{H_2o}} = 45.348\ kJ/kmol \cdot K$

$C_{P_{N_2}} = 34.226\ kJ/kmol \cdot K$

6

$T_{ad,guess} = 2350.30\ K$

$T_{avg} = 1324.15\ K$

$T_{ad,calc} = 2350.33\ K$

$C_{P_{CO_2}} = 57.152\ kJ/kmol \cdot K$

$C_{P_{H_2o}} = 45.292\ kJ/kmol \cdot K$

$C_{P_{N_2}} = 34.207\ kJ/kmol \cdot K$

Calculate new Tad:

$$T_{ad} = 298.15\ K + \frac{-249950\ - (-5324540) + 0\ [kJ/kmol]}{(8 \times 57.152 + 9 \times 45.292 + 47 \times 34.207)[kJ/kmol]} = 2350.33$$

$$\mathrm{T_{ad,guess}} \approx \mathrm{T_{ad,new}} \qquad 3000\ \mathrm{K} \gg 2350.33\ \mathrm{K}$$

Method 3: Cp_i functions

Requires to know the constants which dictate the change for a certain compound as a function of temperature

As the previous method we will approximate the sensible enthalpy term of the products in order to obtain T_{ad}.

$$\sum N_p(\bar{h}^{\circ}_{f_P} + \int_{T^{\circ}}^{T_{ad}} \overline{Cp}_p \mathrm{dT}) = \sum N_R(\bar{h}^{\circ}_{f_R} + \int_{T^{\circ}}^{T_R} \overline{Cp}_R \mathrm{dT})$$

Where

$$\overline{Cp}_i = a_i + b_i T + c_i T^2 + d_i T^3$$

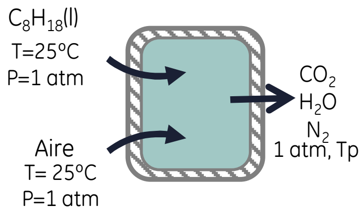
We will look to solve for our unknown variable, Tad.

$$\boxed{\mathrm{T}_{ad} = \frac{\sum N_R(\bar{h}^{\circ}_{f_R} + \int_{T^{\circ}}^{T_R} \overline{Cp}_R \mathrm{dT}) - \left[\sum N_P(\bar{h}^{\circ}_{f_P} - \overline{Cp}_p T^{\circ}\right]}{\sum N_p \overline{Cp}_p T_{guess}}}$$

As with the previous methods, this method will require us to make a supposition or first guess for the value of Tad.

Estimate the adiabatic flame temperature at constant pressure for a stoichiometric combustion of octane with air, if the reactants are at 298K, 1 atm

Suppositions: 1) open system, steady-state; 2)No works is done; 3) All gases are ideal gases



Chemical reaction balanced:

$$C_8H_{18} + 12.5(O_2 + 3.76N_2) \rightarrow 8CO_2 + 9H_2O + 47N_2$$

Based on the energy balance:

$$\sum N_P(\bar{h}_f^\circ)_P + \int_{T^\circ}^{T_{ad}} \overline{Cp}_p \mathrm{dT}) = \sum N_R(\bar{h}_f^\circ)_R + \int_{T^\circ}^{T_R} \overline{Cp}_R \mathrm{dT})$$

$$\sum \mathrm{N_R}(\bar{\mathrm{h}}_\mathrm{f}^\circ)_\mathrm{R} = -249950 \text{ kJ/kmol}$$

$$\sum \mathrm{N_P}(\bar{\mathrm{h}}_\mathrm{f}^\circ)_\mathrm{P} = -5\,324\,540 \text{ kJ/kmol}$$

$$\sum \mathrm{N_R} \int_{\mathrm{T}^\circ}^{\mathrm{T_R}} \mathrm{Cp_R dT} = 0 \text{ kJ/kmol}$$

The sensible enthalpy term of the products can be interpreted:

$$\sum N_p (\int_{T^\circ}^{T_{ad}} \overline{Cp}_p \mathrm{dT}) = \sum N_p\,(\overline{Cp}_p T_{ad} - \overline{Cp}_p T^\circ)$$

Calculating Cp_p @ T:

1. Suppose a T_{ad} value $\qquad \mathrm{T_{ad,guess}} = 3\,000°\mathrm{K}$
2. Obtain T_{cp} $\qquad \overline{\mathrm{T}}_{cp} = 0.5(3000 + 298) = 1649 \text{ K}$
3. Obtain Cp values @ T_{cp}
   Using the Cp equations

	Ai	Bi	Ci	Di
CO_2	22.26	5.9810E-02	-3.5010E-05	7.4690E-09
H_2O	32.24	1.9230E-03	1.0550E-05	-3.5950E-09
N_2	28.9	-1.5710E-03	8.0810E-06	-2.8730E-09

$$\overline{\mathrm{Cp}}_{\mathrm{CO_2,1649K}} = 59.1782\ \mathrm{kJ/kmol \cdot K}$$
$$\overline{\mathrm{Cp}}_{\mathrm{H_2O,1649K}} = 47.9788\ \mathrm{kJ/kmol \cdot K}$$
$$\overline{\mathrm{Cp}}_{\mathrm{N_2,1649K}} = 35.4009\ \mathrm{kJ/kmol \cdot K}$$

$$\sum N_p \left(\int_{T^\circ}^{T_{ad}} \overline{Cp}_p \mathrm{dT}\right) = \sum N_p \left(\overline{Cp}_p T_{ad} - \overline{Cp}_p T^\circ\right)$$

$$= (N_{CO2}\overline{Cp}_{CO2} + N_{H2O}\overline{Cp}_{H2O} + N_{N2}\overline{Cp}_{N2})T_{ad} - (N_{CO2}\overline{Cp}_{CO2} + N_{H2O}\overline{Cp}_{H2O} + N_{N2}\overline{Cp}_{N2})T^\circ$$

The values of Cp calculated previously are the ones used for this calculations, and we solve for Tad:

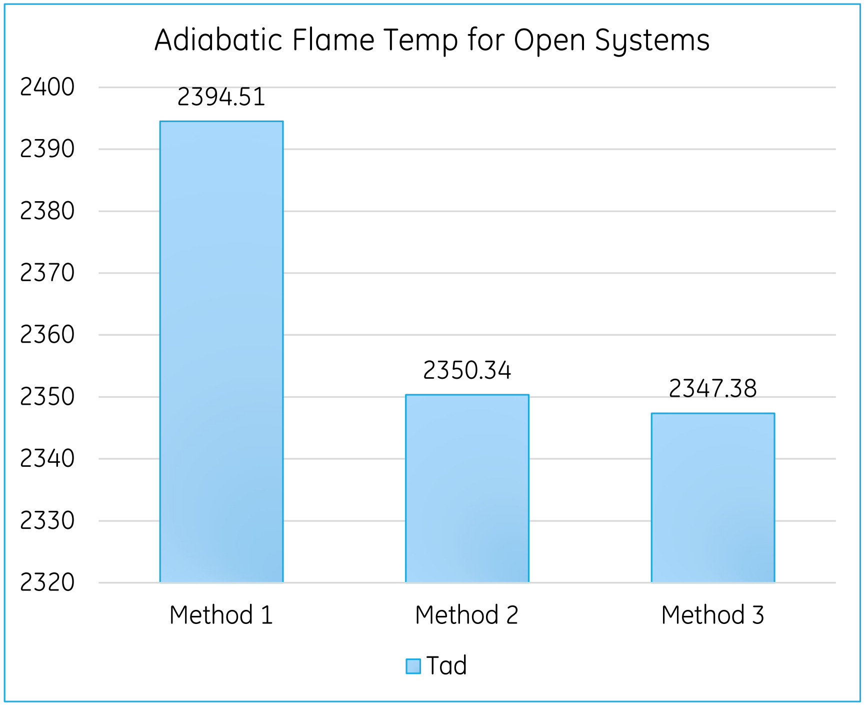
Tad = 2273.26 K << Tguess = 3000 K

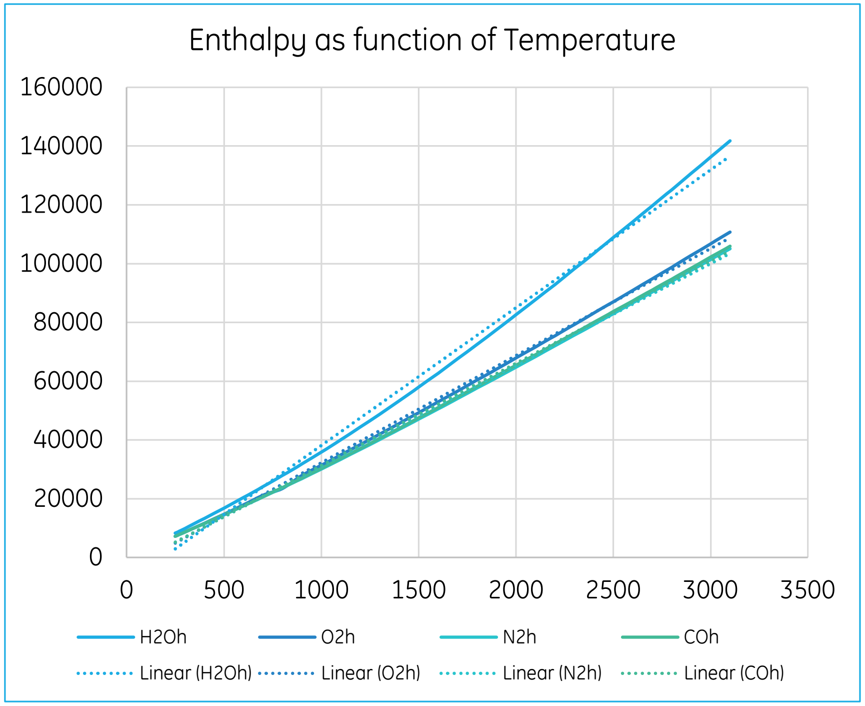
The calculated Tad of 2273K becomes our new guessed temperature
And we repeat the process until the value stabilizes

iteration	Tguess	Tcp	Tad	Diff (K)
1	3000	1649	2273.26	-726.74
2	2273.26	1285.62	2359.11	85.85
3	2359.11	1328.55	2345.57	-13.53
4	2345.57	1321.78	2347.65	2.07
5	2347.65	1322.85	2347.33	-0.31
6	2347.33	1322.66	2347.38	0.049

For this specific process the adiabatic flame temperature is 2 347.3 K

Method Comparison





Determining Tad – Closed systems

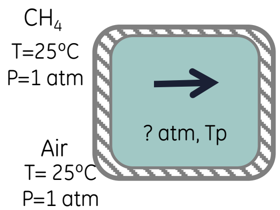
All mentioned methods can be used for closed systems

- Since we suppose all gases behave as ideal gases
- All internal energy and enthalpies depend on temperature only

$$\sum N_P(\bar{h}_f^\circ + \bar{h}(T_{ad}) - \bar{h}^\circ - P\bar{v})_P = \sum N_R(\bar{h}_f^\circ + \bar{h}(T_F) - \bar{h}^\circ - P\bar{v})_R$$

Methane (CH_4) is burned with 300 percent excess air in an adiabatic constant-volume container. Initially, air and methane are at 1 atm and 25°C. Assuming complete combustion, determine the final pressure and temperature of the combustion products.

Suppositions: 1) open system, steady-state; 2)No works is done; 3) All gases are ideal gases



Chemical reaction balanced:

$$CH_4 + 8(O_2 + 3.76N_2) \rightarrow 1CO_2 + 2H_2O + 30.08N_2 + 6O_2$$

Based on the energy balance:

$$\sum N_R(\bar{h}_f^\circ)_R = -74850 \text{ kJ/kmol}$$

$$\sum N_R R_u T_R = 96824 \text{ kJ/kmol}$$

$$\sum N_P(\bar{h}_f^\circ)_P = -877160 \text{ kJ/kmol}$$

$$\sum N_R \int_{T^\circ}^{T_R} Cp_R \, dT = 0 \text{ kJ/kmol}$$

$$T_{ad} = T_0 + \frac{\sum N_R \bar{h}_{f\,R} - \sum N_P \bar{h}^\circ_{f\,P} + \sum N_R\left(\bar{h}(T_F) - \bar{h}^\circ\right)_R}{\sum N_p \overline{Cp}_{avg,p}\left(1 - \frac{R_u}{\overline{Cp}_{avg,p}}\right)}$$

Enthalpy of formation

	h_f°(kJ/kmol)
CH_4	-74850
O_2	0
N_2	0
$H_2O_{(g)}$	-241 820
CO_2	-393 529

The sensible enthalpy term of the products can be interpreted:

$$\sum N_p\left(\int_{T^\circ}^{T_{ad}} \overline{Cp}_p \, dT\right) = \sum N_p\left(\overline{Cp}_p T_{ad} - \overline{Cp}_p T^\circ\right)$$

Calculating Cp_p @ T:

1. Suppose a T_{ad} value
2. Obtain T_{cp}
3. Obtain Cp values @ T_{cp}
   Using the Cp table

$$T_{ad,guess} = 3\,000°K$$

$$\bar{T}_{cp} = 0.5(3000 + 298) = 1649 \text{ K}$$

T	CO2cp	H2Ocp	O2cp	N2cp
298	37.2	33.45	29.31	29.07
300	37.28	33.47	29.33	29.07
400	41.27	34.44	30.21	29.32
500	44.57	35.34	31.11	29.63
600	47.31	36.29	32.03	30.08
700	49.61	37.36	32.93	30.68
800	51.55	38.59	33.76	31.39
900	53.13	39.93	34.45	32.13
1000	54.36	41.31	34.93	32.76

1

$$T_{ad,guess} = 3000\ K$$
$$T_{avg} = 1649\ K$$
$$T_{ad,calc} = 1467.16\ K$$

$$C_{P_{CO_2}} = 59.072\ kJ/kmol \cdot K$$
$$C_{P_{H_2O}} = 48.462\ kJ/kmol \cdot K$$
$$C_{P_{N_2}} = 35.230\ kJ/kmol \cdot K$$
$$C_{P_{O_2}} = 36.900\ kJ/kmol \cdot K$$

3

$$T_{ad,guess} = 1206.67\ K$$
$$T_{avg} = 752.33\ K$$
$$T_{ad,calc} = 1168.60\ K$$

$$C_{P_{CO_2}} = 50.670\ kJ/kmol \cdot K$$
$$C_{P_{H_2O}} = 37.986\ kJ/kmol \cdot K$$
$$C_{P_{N_2}} = 31.042\ kJ/kmol \cdot K$$
$$C_{P_{O_2}} = 33.377\ kJ/kmol \cdot K$$

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$$T_{ad,guess} = 1161.89\ K$$
$$T_{avg} = 729.94\ K$$
$$T_{ad,calc} = 1161.76\ K$$

$$C_{P_{CO_2}} = 50.228\ kJ/kmol \cdot K$$
$$C_{P_{H_2O}} = 37.714\ kJ/kmol \cdot K$$
$$C_{P_{N_2}} = 30.885\ kJ/kmol \cdot K$$
$$C_{P_{O_2}} = 333.189\ kJ/kmol \cdot K$$

Calculate new Tad:

$$T_{ad} = 298.15\ K + \frac{-74850\ - 96824\ - (-877160) + 0 + 377518.1\ [kJ/kmol]}{(1 \times 50.228 + 2 \times 37.714 + 30.08 \times 30.88 + 6 \times 33.189)[kJ/kmol]} = 1161.76$$

$$\mathrm{T_{ad,guess}} \approx \mathrm{T_{ad,new}} \qquad 3000\ \mathrm{K} \gg 1161.76\ \mathrm{K}$$

Method Comparison